

GEMM Analytical Model - Equations

Auto-generated from PYTHON_FORMULAS and LATEX_SYMBOL_OVERRIDES in gemmanalytics.py.

Output order: execution

Machine Parameters (Pre-Defined)

- () : GT Freq (GHz)
- () : XeCU count
- () : XeCore per XeCU
- () : EU per XeCore
- () : L2 banks per XeCU
- () : bank capacity (MB)
- () : DPAS depth
- () : compute efficiency %
- () : L1 read max (B/EU/clock)
- () : L1 write max (B/EU/clock)
- () : GTI read max BW (B/clock)
- () : GTI write max BW (B/clock)
- () : max possible HBM BW (GB/s)

Workload Parameters (Pre-Defined)

- () : input A data format (e.g. fp8)
- () : input B data format (e.g. fp4)
- () : output D data format (e.g. fp8)
- () : M dimension
- () : K dimension
- () : N dimension
- () : output bytes per element (fp32)
- () : machine occupancy %
- () : M per thread
- () : K per thread
- () : N per thread
- () : TG width in units of thread

- () : TG height in units of thread
- () : XeCore cluster width in units of TG (keep cluster size as 4)
- () : XeCore cluster height in units of TG (keep cluster size as 4)
- () : XeCU tile width in units of TG
- () : XeCU tile height in units of TG
- () : GPU tile width in XeCU unit

Computed Equations

row 22 - INPUT A BYTES PER ELEMENT

$$\text{Bytes}_{\text{perElement}}^{(\text{matA})} = \text{DataFormatToBytes}[\text{Fmt}^{(\text{matA})}]$$

row 23 - INPUT B BYTES PER ELEMENT

$$\text{Bytes}_{\text{perElement}}^{(\text{matB})} = \text{DataFormatToBytes}[\text{Fmt}^{(\text{matB})}]$$

row 25 - OUTPUT BYTES PER ELEMENT AFTER DOWN CONVERSION

$$\text{Bytes}_{\text{perElement}}^{(\text{matD}\downarrow)} = \text{DataFormatToBytes}[\text{Fmt}^{(\text{matD}\downarrow)}]$$

row 33 - EU COUNT

$$|\text{EU}| = |\text{XeCore}|_{\text{perXeCU}} \times |\text{EU}|_{\text{perXeCore}} \times |\text{XeCU}|$$

row 36a - MMA MAC THROUGHPUT PER EU

$$\tau_{\text{mMACp}(\text{Clk}\cdot\text{EU})}^{(\text{peak})} = \frac{4}{\max(\text{Bytes}_{\text{perElement}}^{(\text{matA})}, \text{Bytes}_{\text{perElement}}^{(\text{matB})})} \times D_{\text{DPAS}} \times 16$$

row 36 - MMA MAC THROUGHPUT PER XECORE

$$\tau_{\text{mMACp}(\text{Clk}\cdot\text{XeCore})}^{(\text{peak})} = \tau_{\text{mMACp}(\text{Clk}\cdot\text{EU})}^{(\text{peak})} \times |\text{EU}|_{\text{perXeCore}}$$

row 42 - CLKS PER DPAS

$$\text{CLKS}_{\text{DPAS}} = \frac{M_{\text{perThread}} \times K_{\text{perThread}} \times N_{\text{perThread}}}{\frac{\tau_{\text{mMACp}(\text{Clk}\cdot\text{XeCore})}^{(\text{peak})}}{|\text{EU}|_{\text{perXeCore}}}}$$

row 43 - THREAD WIDTH IN UNITS OF ELEMENTS

$$W_{\text{element}}^{(\text{Thread})} = N_{\text{perThread}}$$

row 44 - THREAD HEIGHT IN UNITS OF ELEMENTS

$$H_{\text{element}}^{(\text{Thread})} = M_{\text{perThread}}$$

row 47 - TG WIDTH IN UNITS OF ELEMENT REALIZED BY MULTIPLE MMA ITERATIONS

$$W_{\text{element}}^{(\text{ThreadGroup, realized by multiple MMA iterations})} = W_{\text{thread}}^{(\text{ThreadGroup})} \times W_{\text{element}}^{(\text{Thread})}$$

row 14 - TG TILES IN N

$$|\text{Tiles}|_{\text{N}}^{(\text{TG})} = \frac{N_{\text{dim}}}{W_{\text{element}}^{(\text{ThreadGroup, realized by multiple MMA iterations})}}$$

row 48 - TG HEIGHT IN UNITS OF ELEMENT

$$H_{\text{element}}^{(\text{ThreadGroup})} = H_{\text{thread}}^{(\text{ThreadGroup})} \times H_{\text{element}}^{(\text{Thread})}$$

row 15 - TG TILES IN M

$$|\text{Tiles}|_{\text{M}}^{(\text{TG})} = \frac{M_{\text{dim}}}{H_{\text{element}}^{(\text{ThreadGroup})}}$$

row 52 - XECORE CLUSTER WIDTH IN UNITS OF ELEMENT

$$W_{\text{element}}^{(\text{XeCoreCluster})} = W_{\text{TG, keep cluster size as 4}}^{(\text{XeCoreCluster})} \times W_{\text{element}}^{(\text{ThreadGroup, realized by multiple MMA iterations})}$$

row 16 - TG CLUSTER TILES IN N

$$|\text{Tiles}|_{\text{N}}^{(\text{TG Cluster})} = \frac{N_{\text{dim}}}{W_{\text{element}}^{(\text{XeCoreCluster})}}$$

row 53 - XECORE CLUSTER HEIGHT IN UNITS OF ELEMENT

$$H_{\text{element}}^{(\text{XeCoreCluster})} = H_{\text{TG, keep cluster size as 4}}^{(\text{XeCoreCluster})} \times H_{\text{element}}^{(\text{ThreadGroup})}$$

row 17 - TG CLUSTER TILES IN M

$$|\text{Tiles}|_{\text{M}}^{(\text{TG Cluster})} = \frac{M_{\text{dim}}}{H_{\text{element}}^{(\text{XeCoreCluster})}}$$

row 56 - XECU TILE WIDTH IN UNITS OF ELEMENT

$$W_{\text{element}}^{(\text{XeCUTile})} = W_{\text{TG}}^{(\text{XECUTile})} \times W_{\text{element}}^{(\text{ThreadGroup, realized by multiple MMA iterations})}$$

row 18 - XECU TILES IN N

$$|\text{Tiles}|_{\text{N}}^{(\text{XeCU})} = \frac{N_{\text{dim}}}{W_{\text{element}}^{(\text{XeCUTile})}}$$

row 20 - GPU TILES IN N

$$|\text{Tiles}|_{\text{N}}^{(\text{GPU})} = \frac{\frac{N_{\text{dim}}}{W_{\text{element}}^{(\text{XeCUTile})}}}{W_{\text{XeCU}}^{(\text{GPURTILE})}}$$

row 57 - XECU TILE HEIGHT IN UNITS OF ELEMENT

$$H_{\text{element}}^{(\text{XeCUTile})} = H_{\text{TG}}^{(\text{XECUTile})} \times H_{\text{element}}^{(\text{ThreadGroup})}$$

row 19 - XECU TILES IN M

$$|\text{Tiles}|_{\text{M}}^{(\text{XeCU})} = \frac{M_{\text{dim}}}{H_{\text{element}}^{(\text{XeCUTile})}}$$

row 59 - GPU TILE HEIGHT IN XECU UINT

$$H_{\text{XeCU}}^{(\text{GPUTile})} = \frac{|\text{XeCU}|}{W_{\text{XeCU}}^{(\text{GPUTile})}}$$

row 21 - GPU TILES IN M

$$|\text{Tiles}|_{\text{M}}^{(\text{GPU})} = \frac{\frac{M_{\text{dim}}}{H_{\text{element}}^{(\text{XeCUTile})}}}{H_{\text{XeCU}}^{(\text{GPUTile})}}$$

row 13 - WAVES

$$N_{\text{waves}} = \text{CEILING}(|\text{Tiles}|_{\text{N}}^{(\text{GPU})}, 1) \times \text{CEILING}(|\text{Tiles}|_{\text{M}}^{(\text{GPU})}, 1)$$

row 60 - GPU TILE WIDTH IN UNITS OF ELEMETNS

$$W_{\text{element}}^{(\text{GPUTile})} = W_{\text{XeCU}}^{(\text{GPUTile})} \times W_{\text{element}}^{(\text{XeCUTile})}$$

row 61 - GPU TILE HEIGHT IN UNITS OF ELEMETNS

$$H_{\text{element}}^{(\text{GPUTile})} = H_{\text{XeCU}}^{(\text{GPUTile})} \times H_{\text{element}}^{(\text{XeCUTile})}$$

row 63 - MAT A INPUT SIZE B

$$\text{Size}_{\text{B}}^{(\text{matA})} = M_{\text{dim}} \times K_{\text{dim}} \times \text{Bytes}_{\text{perElement}}^{(\text{matA})}$$

row 64 - MAT B INPUT SIZE B

$$\text{Size}_{\text{B}}^{(\text{matB})} = K_{\text{dim}} \times N_{\text{dim}} \times \text{Bytes}_{\text{perElement}}^{(\text{matB})}$$

row 65 - MAT C INPUT D OUTPUT SIZE B

$$\text{Size}_{\text{B}}^{(\text{matC}, \text{matD})} = M_{\text{dim}} \times N_{\text{dim}} \times \text{Bytes}_{\text{perElement}}^{(\text{matD}\downarrow)}$$

row 66 - MAT D INTERMEDIATE SIZE B

$$\text{Size}_{\text{B}}^{(\text{matD}\downarrow)} = M_{\text{dim}} \times N_{\text{dim}} \times \text{Byte}_{\text{perElement}}^{(\text{matD})}$$

row 102 - TOTAL L2 SIZE B FOR A SINGLE INSTANCE

$$\text{L2Size}_{\text{B}} = \text{L2BankSize}_{\text{MB}} \times |\text{L2Banks}|_{\text{perXeCU}} \times 1024 \times 1024$$

**row 103 - WORKING DATA SET SIZE OF K IN L2 CORRESP 20K CLOCKS OF
THREAD DIVERGENCE**

$$|\text{WorkingSet}|_{20\text{K clks of thread divergence}}^{(\text{K in L2})} = \min(20000 \times \frac{\text{K}_{\text{perThread}}}{\text{CLKS}_{\text{DPAS}}}, \text{K}_{\text{dim}})$$

**row 104 - TOTAL REQUIRED L2 SIZE FOR IDEAL HIT RATE B FOR A SINGLE
INSTANCE AND SINGLE WAVE**

$$\begin{aligned} \text{TotalRequiredL2Size}_{\text{matB always hit}}^{1 \text{ instance, 1 wave}} &= \text{H}_{\text{element}}^{(\text{XeCUTile})} \times \text{Bytes}_{\text{perElement}}^{(\text{matA})} \times |\text{WorkingSet}|_{20\text{K clks of thread divergence}}^{(\text{K in L2})} \\ &+ \text{W}_{\text{element}}^{(\text{XeCUTile})} \times \text{Bytes}_{\text{perElement}}^{(\text{matB})} \times |\text{WorkingSet}|_{20\text{K clks of thread divergence}}^{(\text{K in L2})} \\ &+ \text{W}_{\text{element}}^{(\text{XeCUTile})} \times \text{H}_{\text{element}}^{(\text{XeCUTile})} \times \text{Bytes}_{\text{perElement}}^{(\text{matD})} \end{aligned}$$

row 125 - MAX POSSIBLE HBM BW FREQ B CLK

$$\text{MemBW}_{\text{BpClk}}^{\text{max}} = \frac{\text{MemBW}_{\text{GBps}}^{\text{max}}}{f_{\text{GHz}}^{(\text{GT})}}$$

row 37a - WORKLOAD MAC PER XECORE

$$\text{WL}_{\text{MAC}}^{(\text{XeCore})} = \frac{\text{M}_{\text{dim}} \times \text{K}_{\text{dim}} \times \text{N}_{\text{dim}}}{|\text{XeCore}|_{\text{perXeCU}} \times |\text{XeCU}|}$$

row 37 - CLK SPECIFIED EFFICIENCY

$$\text{T}_{\text{clk}}^{(\text{total})} = \frac{\text{WL}_{\text{MAC}}^{(\text{XeCore})}}{\tau_{\text{mMACp}(\text{Clk} \cdot \text{XeCore})}^{(\text{peak})} \times \eta_{\text{systolic}}}$$

row 69 - TOTAL L2 READ B

$$\begin{aligned} \text{L2Rd}_{\text{B}}^{(\text{total})} &= \text{Size}_{\text{B}}^{(\text{matA})} \times \text{CEILING}\left(\frac{\text{N}_{\text{dim}}}{\text{W}_{\text{element}}^{(\text{ThreadGroup, realized by multiple MMA iterations})}}, 1\right) \\ &+ \text{Size}_{\text{B}}^{(\text{matB})} \times \text{CEILING}\left(\frac{\text{M}_{\text{dim}}}{\text{H}_{\text{element}}^{(\text{ThreadGroup})}}, 1\right) \end{aligned}$$

row 70 - TOTAL L2 WRITE B

$$\text{L2Wr}_{\text{B}}^{(\text{total})} = \text{Size}_{\text{B}}^{(\text{matC, matD})}$$

row 71 - TOTAL L1 READ B

$$\begin{aligned} \text{L1Rd}_{\text{B}}^{(\text{total})} &= \text{Size}_{\text{B}}^{(\text{matA})} \times \text{CEILING}\left(\frac{\text{N}_{\text{dim}}}{\text{W}_{\text{element}}^{(\text{Thread})}}, 1\right) \\ &+ \text{Size}_{\text{B}}^{(\text{matB})} \times \text{CEILING}\left(\frac{\text{M}_{\text{dim}}}{\text{H}_{\text{element}}^{(\text{Thread})}}, 1\right) \end{aligned}$$

row 72 - TOTAL L1 WRITE B

$$L1Wr_B^{(total)} = Size_B^{(matC, matD)}$$

row 74 - L2 READ B XECORE CLK

$$L2RdBW_{Bp(Clk \cdot XeCore)} = \frac{\frac{L2Rd_B^{(total)}}{|XeCU| \times |XeCore|_{perXeCU}}}{T_{clk}^{(total)}}$$

row 75 - L2 WRITE B XECORE CLK

$$L2WrBW_{Bp(Clk \cdot XeCore)} = \frac{\frac{L2Wr_B^{(total)}}{|XeCU| \times |XeCore|_{perXeCU}}}{T_{clk}^{(total)}}$$

row 76 - L2 READ WRITE B XECORE CLK

$$L2RdWrBW_{Bp(Clk \cdot XeCore)} = L2RdBW_{Bp(Clk \cdot XeCore)} + L2WrBW_{Bp(Clk \cdot XeCore)}$$

row 79 - L1 READ B EU CLK

$$L1RdBW_{Bp(Clk \cdot EU)} = \frac{\frac{L1Rd_B^{(total)}}{|EU|}}{T_{clk}^{(total)}}$$

row 77 - L1 READ B XECORE CLK

$$L1RdBW_{Bp(Clk \cdot XeCore)} = L1RdBW_{Bp(Clk \cdot EU)} \times |EU|_{perXeCore}$$

row 80 - L1 WRITE B EU CLK

$$L1WrBW_{Bp(Clk \cdot EU)} = \frac{\frac{L1Wr_B^{(total)}}{|EU|}}{T_{clk}^{(total)}}$$

row 78 - L1 WRITE B XECORECLK

$$L1WrBW_{Bp(Clk \cdot XeCore)} = L1WrBW_{Bp(Clk \cdot EU)} \times |EU|_{perXeCore}$$

row 83 - L2 READ MAX B XECORE CLK

$$L2RdBW_{Bp(Clk \cdot XeCore)}^{(max)} = \frac{|XeCore|_{perXeCU} \times 64}{|XeCore|_{perXeCU}}$$

row 84 - L2 WRITE MAX B XECORE CLK

$$L2WrBW_{Bp(Clk \cdot XeCore)}^{(max)} = L2RdBW_{Bp(Clk \cdot XeCore)}^{(max)}$$

row 85 - L2 READ WRITE MAX B XECORE CLK

$$L2RdWrBW_{Bp(Clk \cdot XeCore)}^{(max)} = L2WrBW_{Bp(Clk \cdot XeCore)}^{(max)}$$

row 90 - L2 READ B XECORE CLK PCT

$$\eta_{L2RdBWpXeCore} = \frac{L2RdBW_{Bp(Clk \cdot XeCore)}}{L2RdBW_{Bp(Clk \cdot XeCore)}^{(max)}}$$

row 91 - L2 WRITE B XECORE CLK PCT

$$\eta_{L2WrBWpXeCore} = \frac{L2WrBW_{Bp(Clk \cdot XeCore)}}{L2WrBW_{Bp(Clk \cdot XeCore)}^{(max)}}$$

row 92 - L2 READ WRITE B XECORE CLK PCT

$$\eta_{L2RdWrBWpXeCore} = \frac{L2RdWrBW_{Bp(Clk \cdot XeCore)}}{L2RdWrBW_{Bp(Clk \cdot XeCore)}^{(max)}}$$

row 93 - L1 READ B EU CLK PCT

$$\eta_{L1RdBWpEU} = \frac{L1RdBW_{Bp(Clk \cdot EU)}}{L1RdBW_{Bp(Clk \cdot EU)}^{max}}$$

row 94 - L1 WRITE B EU CLK PCT

$$\eta_{L1WrBWpEU} = \frac{L1WrBW_{Bp(Clk \cdot EU)}}{L1WrBW_{Bp(Clk \cdot EU)}^{max}}$$

row 106 - L2 HIT RATE ASSUMED RANDOM ACCESS WITHIN THE WORKING DATA SET PCT

$$L2HitRate_{random \text{ WS access}} = \min\left(\frac{L2Size_B}{TotalRequiredL2Size_{matB \text{ alwayshit}}^{1 \text{ instance, 1 wave}}}, 1\right)$$

row 107 - L2 MISS RATE PCT

$$L2MissRate = 1 - L2HitRate_{random \text{ WS access}}$$

row 108 - TOTAL L2 READ TRAFFIC B

$$L2RdB_{total} = L2RdB_B^{(total)}$$

row 110 - PROBABILITY OF MATA HIT IN L2 DURING A NON FIRST WAVE

PCT

$$P_{L2Hit, after 1st wave}^{(matA)} = IF(Size_B^{(matA)} + Size_B^{(matB)} + Size_B^{(matC, matD)} \leq L2Size_B, 1.0, IF(K_{dim} > 2 \times |WorkingSet|_{20K \text{ clks of thread divergence}}^{(K \text{ in L2})}, 0.0, 1 - \frac{K_{dim} - |WorkingSet|_{20K \text{ clks of thread divergence}}^{(K \text{ in L2})}}{|WorkingSet|_{20K \text{ clks of thread divergence}}^{(K \text{ in L2})}}))$$

row 111 - PROBABILITY OF MATB HIT IN L2 DURING A NON FIRST WAVE

PCT

$$P_{L2Hit, after 1st wave}^{(matB)} = IF(Size_B^{(matA)} + Size_B^{(matB)} + Size_B^{(matC, matD)} \leq L2Size_B, 1.0, 0.0)$$

row 112 - PROBABILITY OF MATA MISS IN L2 DURING A NON FIRST WAVE

PCT

$$P_{L2Miss, after 1st wave}^{(matA)} = 1 - P_{L2Hit, after 1st wave}^{(matA)}$$

row 113 - PROBABILITY OF MATB MISS IN L2 DURING A NON FIRST WAVE

PCT

$$P_{L2Miss, after 1st wave}^{(matB)} = 1 - P_{L2Hit, after 1st wave}^{(matB)}$$

row 116 - TOTAL HBM READ B AFTER A COMPLETION OF A WAVE CONSIDER

COLD CACHE

$$MemRdB_{cold start}^{(wave)} = \frac{\left(MemRdB_{cold start}^{(wave)} \right)_{num}}{2}$$

$$\begin{aligned}
\left(\text{MemRdB}_{\text{cold start}}^{(\text{wave})}\right)_{\text{num}} &= \text{Size}_B^{(\text{matA})} \\
&+ \text{Size}_B^{(\text{matA})} \times \left(\text{CEILING}\left(\frac{N_{\text{dim}}}{W_{\text{element}}^{(\text{XeCUTile})}}, 1\right) - 1 \right) \times P_{\text{L2Miss, after 1st wave}}^{(\text{matA})} \\
&+ \text{Size}_B^{(\text{matB})} \\
&+ \text{Size}_B^{(\text{matB})} \times \left(\text{CEILING}\left(\frac{M_{\text{dim}}}{H_{\text{element}}^{(\text{XeCUTile})}}, 1\right) - 1 \right) \times P_{\text{L2Miss, after 1st wave}}^{(\text{matB})} \\
&+ \left(\left(\text{MemRdB}_{\text{cold start}}^{(\text{wave})}\right)_{\text{num}} \right)_{\text{aux4}} \times \text{L2MissRate}
\end{aligned}$$

$$\begin{aligned}
\left(\left(\text{MemRdB}_{\text{cold start}}^{(\text{wave})}\right)_{\text{num}} \right)_{\text{aux4}} &= \text{L2RdB}_{\text{total}} \\
&- \left(\left(\left(\text{MemRdB}_{\text{cold start}}^{(\text{wave})}\right)_{\text{num}} \right)_{\text{aux4}} \right)_{\text{aux1}}
\end{aligned}$$

$$\begin{aligned}
\left(\left(\left(\text{MemRdB}_{\text{cold start}}^{(\text{wave})}\right)_{\text{num}} \right)_{\text{aux4}} \right)_{\text{aux1}} &= \text{Size}_B^{(\text{matA})} \\
&+ \text{Size}_B^{(\text{matA})} \times \left(\text{CEILING}\left(\frac{N_{\text{dim}}}{W_{\text{element}}^{(\text{XeCUTile})}}, 1\right) - 1 \right) \times P_{\text{L2Miss, after 1st wave}}^{(\text{matA})} \\
&+ \text{Size}_B^{(\text{matB})} \\
&+ \text{Size}_B^{(\text{matB})} \times \left(\text{CEILING}\left(\frac{M_{\text{dim}}}{H_{\text{element}}^{(\text{XeCUTile})}}, 1\right) - 1 \right) \times P_{\text{L2Miss, after 1st wave}}^{(\text{matB})}
\end{aligned}$$

row 117 - TOTAL HBM WRITE B

$$\text{MemWrB}^{(\text{total})} = \text{Size}_B^{(\text{matC,matD})}$$

row 118 - TOTAL HBM B

$$\text{MemRdWrB}^{(\text{total})} = \text{MemWrB}^{(\text{total})} + \text{MemRdB}_{\text{cold start}}^{(\text{wave})}$$

row 121 - HBM READ B CLK

$$\text{MemRdBW}_{\text{BpClk}} = \frac{\text{MemRdB}_{\text{cold start}}^{(\text{wave})}}{T_{\text{clk}}^{(\text{total})}}$$

row 122 - HBM WRITE B CLK

$$\text{MemWrBW}_{\text{BpClk}} = \frac{\text{MemWrB}^{(\text{total})}}{T_{\text{clk}}^{(\text{total})}}$$

row 123 - HBM TOTAL B CLK

$$\text{MemRdWrBW}_{\text{BpClk}} = \frac{\text{MemRdWrB}^{(\text{total})}}{T_{\text{clk}}^{(\text{total})}}$$

row 126 - HBM BW PCT

$$\eta_{\text{MemBW}} = \frac{\text{MemRdWrBW}_{\text{BpClk}}}{\text{MemBW}_{\text{BpClk}}^{\text{max}}}$$

row 127 - GTI READ BW PCT

$$\eta_{\text{GTIRdBW}} = \frac{\text{MemRdBW}_{\text{BpClk}}}{\text{GtiRdBW}_{\text{BpClk}}^{\text{max}}}$$

row 128 - GTI WRITE BW PCT

$$\eta_{\text{GTIW_rBW}} = \frac{\text{MemWrBW}_{\text{BpClk}}}{\text{GtiWrBW}_{\text{BpClk}}^{\text{max}}}$$